

Changes made in response to your comments of 19 September

This page lists each comment and what changed. The last page shows the edited text with every change marked: **red = removed**, **blue = added**. Only the abstract and the first paragraph of the Introduction changed, so only those are shown. The body is still four pages and no number changed.

1. **The abstract’s opening list reads as the only things a light curve encodes.**

Now: “TESS light curves encode a star’s pulsation, rotation, binarity and oscillations, **among other physical properties**, but the survey produces far more light curves than can be labelled by hand.”

2. **Say exactly which parameters we work on.**

Was: “...on ten tasks **spanning variability, asteroseismology and rotation**.”

Now: “...on ten tasks: **five variability classes, four asteroseismic targets and rotation period**.”

The full list of ten did not fit in the four-page limit; it is in Table 1. One nuance: pre-training uses no labels, so these are the quantities we *evaluate* on, not train on.

3. **“Why do we care” about those quantities is never set up.**

Not changed in this version. The four-page limit leaves room for one clause in the Introduction (“and with them its age, interior and companions”). A motivation paragraph is on the list for the arXiv version.

4. **Cadence language is too fixed (full-frame images are now 200 s).**

Was: “full-frame images, **every 30 or 10 minutes**, for every star in the field”.

Now: “full-frame images of every star in the field **at cadences that change between mission cycles (30 minutes, then 10, now 200 seconds)**”.

5. **TESS does not trade sampling rate for sky coverage; it trades spatial resolution.**

The sentence was wrong as written and is removed. I had meant that, within TESS, only pre-selected targets get 2-minute data; it read as a TESS-versus-Kepler claim.

Was: “TESS **trades sampling rate against sky coverage**: 2-minute (or 20-second) light curves for...”

Now: “TESS **returns** 2-minute or 20-second light curves for...”

The pixel-scale and contamination point is not in this version (it needs a citation and space). It is on the arXiv list, where it also bears on label noise in our catalogue cross-matches.

6. **“LSST will widen the gap” — which gap?**

Was: “LSST **will widen the gap**.”

Now, with your wording: “LSST **will add billions more light curves, albeit at a cadence of days**.”

7. **Figure 1: hard to tell which two points belong to which row.**

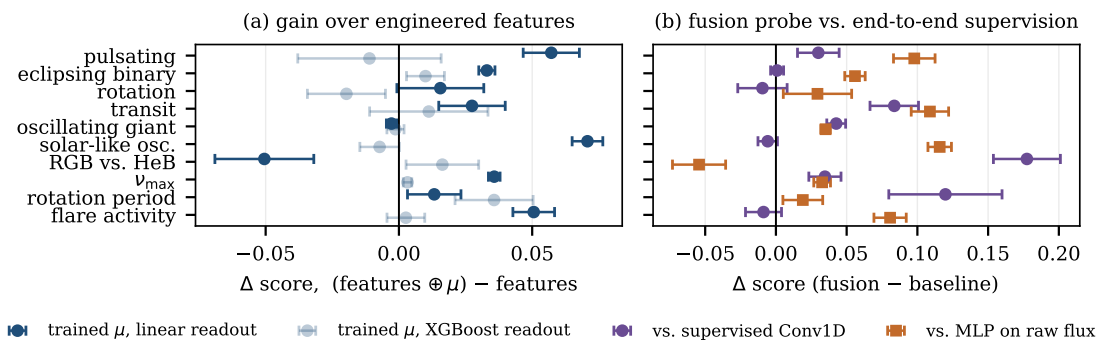
Alternate rows are now shaded, a faint dotted line runs through each row, and the two points of each row sit closer together, so each pair traces back to its task label.

The two versions are on the next page.

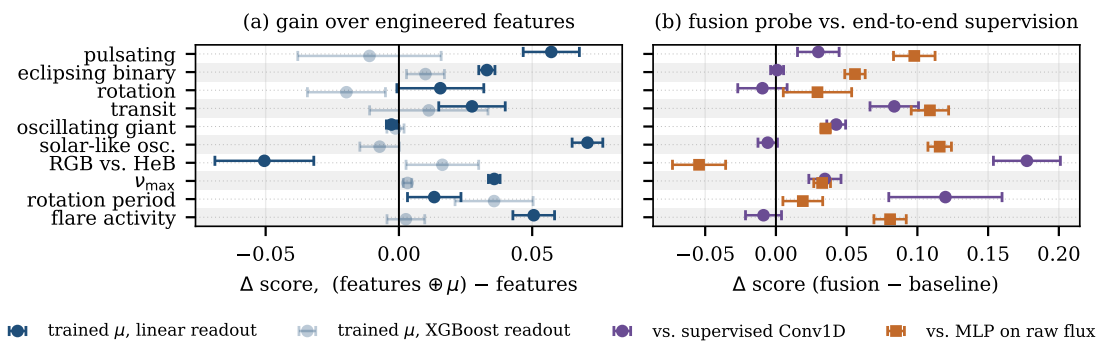
Small trims made only to hold the four-page limit (each of the edits above added a line): “features from **these light curves**” in the abstract; “Wide-field **photometric** surveys” and “a cheap readout for each **downstream** task” in the Introduction; “**A self-supervised** encoder here therefore has to be chosen by a held-out probe” in the Discussion (not shown on the last page).

Figure 1, before and after

Before



After



What does self-supervision add to engineered features for stellar light curves?

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Abstract

TESS light curves encode a star’s pulsation, rotation, binarity and oscillations, among other physical properties, but the survey produces far more light curves than can be labelled by hand. Astronomers have engineered a few dozen physically motivated features from these light curves, which together remain a strong baseline. Previous work has trained encoders on light curves and judged them by their own downstream scores, but has not asked what they add to the engineered features. We pre-train a convolutional encoder on unlabelled TESS light curves to predict each window’s latent representation from those of its neighbours. We then freeze it, add its features to the 25 engineered features, and read the combination out with one fixed linear probe on ten tasks spanning variability, asteroseismology and rotation: five variability classes, four asteroseismic targets and rotation period. Adding the learned features improves the readout on 7 of 10 tasks over the engineered features alone, while an untrained encoder does not. The combination also outperforms two networks trained end-to-end on the same labelled stars on most tasks. An encoder trained without labels therefore gives astronomers new features that complement the ones they already use rather than replace them.

1 Introduction

A star’s light curve carries its pulsation, rotation, eclipses and oscillations, and with them its age, interior and companions. Wide-field photometric surveys now produce light curves far faster than they can be labelled. TESS [Ricker et al., 2015] trades sampling rate against sky coverage: returns 2-minute (or 20-second) [Huber et al., 2022] light curves for hundreds of thousands of pre-selected stars, and full-frame images every 30 or 10 minutes, for every of every star in the field [Huber et al., 2022] at cadences that change between mission cycles (30 minutes, then 10, now 200 seconds). LSST [Ivezić et al., 2019] will widen the gap add billions more light curves, albeit at a cadence of days. Self-supervised learning is the natural response: pre-train one encoder on the unlabelled archive, then attach a cheap readout for each downstream task. Encoders of this shape exist for ground-based, Gaia and Kepler light curves [Donoso-Oliva et al., 2025, Zuo et al., 2025, Rui, 2026] and are evaluated by classification accuracy, sometimes against engineered features [Rui, 2026]; a label-free TESS representation [Poznanski, 2026] is evaluated by whether repeat observations of one star lie close in its embedding space. None measures what they add to engineered features.